

HIMANK SINGHVI

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Education

Indian Institute of Information Technology (IIIT), Hyderabad

Aug 2023 – May 2027 (Expected)

B.Tech in Computer Science and Engineering / CGPA: 7.58/10

Hyderabad, India

- **Relevant Coursework:** Data Structures and Algorithms, Operating Systems and Networks, Computer Systems Organization, Data and Applications (DBMS), Software Systems Design, Statistical Methods in AI, Machine Learning.

Technical Skills

Languages: Python, Java, C, C++, JavaScript, SQL, Shell Scripting (Bash), HTML, CSS

Backend & Web: Spring Boot, Node.js, Flask, REST APIs, Microservices, Next.js, React Native, Tailwind CSS

AI / ML: PyTorch, NumPy, Transformers, LSTM/RNN, Reinforcement Learning, RAG, Vector Embeddings, Semantic Search, LLM Reranking

Cloud & DevOps: AWS (EC2, RDS), Azure Kubernetes Service (AKS), Kubernetes, Jenkins, CI/CD, Kafka, Git, n8n

Databases: PostgreSQL, MySQL, MongoDB

Core CS & Robotics: Data Structures and Algorithms, Operating Systems, Computer Networks, Concurrency, Distributed Systems, ROS2, Gazebo

Experience

Chubb

Jun 2026 – Jul 2026

Software Development Engineer Intern

- Modernized the legacy “Third Party” Java application into a Spring Boot microservice deployed on Azure Kubernetes Service (AKS).
- Reduced manual release effort by building an automated Jenkins CI/CD pipeline covering build, test, and deployment stages.
- Improved inter-service data-flow reliability by configuring Dynaflow to orchestrate the pipeline build.

Zigsaw | zigsaw.in

2026

Collaborator under [Vaibhav Chouhan](#)

- Architected a Retrieval-Augmented Generation (RAG) pipeline to automate candidate screening over large resume datasets.
- Used semantic search with vector embeddings to retrieve the top 100 resumes by job-description similarity.
- Implemented an LLM-based listwise reranking agent to refine the shortlist to the top 10 candidates.

BoardGameiac

2025

Software Developer Intern under [Kartik Sanghavi](#)

- Developed a full-stack web application: frontend, admin dashboard, backend APIs, and role-based user permission controls.
- Deployed on AWS using EC2 for the application server and RDS for the PostgreSQL database.
- Integrated the application with the Petpooja point-of-sale platform.

Future Leaders

2024

Software Developer Intern under [Aparna Renganathan](#)

- Designed the frontend (admin page, feed, gamification) and built backend APIs and database schema for user management and content delivery.

Projects

Autonomous Target-Chasing Drone | *ROS2, Gazebo, Reinforcement Learning*

Project Lead under [Harikumar Kandath](#)

- Leading a team building a LiDAR, camera, and radar equipped drone that autonomously tracks and chases a target drone in a ROS2/Gazebo simulation.
- Implemented vision-based target detection and a Reinforcement Learning pursuit policy; managing component integration and system planning.

Spatiotemporal Flood Forecasting with LSTM | *Python, PyTorch, NumPy, Time-Series*

- Developed an LSTM hydrological model to forecast water levels and flood risk in urban river basins (e.g., Musi River Basin) from historical multivariate climate data.
- Integrated IMD climate datasets with Digital Elevation Model (DEM) terrain features to build spatiotemporal input sequences.
- Chose LSTMs over MLPs/vanilla RNNs: gated cell state (forget, input, output) mitigates vanishing gradients in BPTT and retains long-range dependencies.

Transformer Model from Scratch | *Python, NumPy, PyTorch*

- Implemented multi-head self-attention, positional encoding, and layer normalization from first principles to build an encoder-decoder Transformer.
- Built a custom automatic differentiation engine and training loop; validated gradient correctness and convergence on a sequence-to-sequence task.

Network File System (NFS) | *C, Networking, Concurrency*

- Built a scalable, fault-tolerant distributed file system supporting file and metadata operations, with exclusive-access concurrency control, mutex locks, Tries, LRU caching, async writes, logging, and real-time audio streaming.

Custom Unix-like Shell | *C, POSIX API*

- Developed a shell supporting built-in commands, aliases, I/O redirection, background/foreground processes, and online man-page lookup.

Leadership and Achievements

- **Head of Logistics, Student Alumni Connect Cell (2026 – Present):** manage logistics for events, talks, and networking sessions between startup founders and students.
- JEE Mains (AIR 426), JEE Advanced (AIR 2717), STSE Class 10 (Rank 95), STSE Class 12 (Rank 8), Dean’s List (Spring 2026)